

19ECE204
Digital Electronics and Systems
III Semester
B. Tech. (ECE & CSE)
Amrita School of Engineering, Bengaluru
Counters

Overview

- Counters
 - Asynchronous Counter
 - Synchronous Counter
- Other Types of Counters
 - Ring Counter
 - Johnson Counter

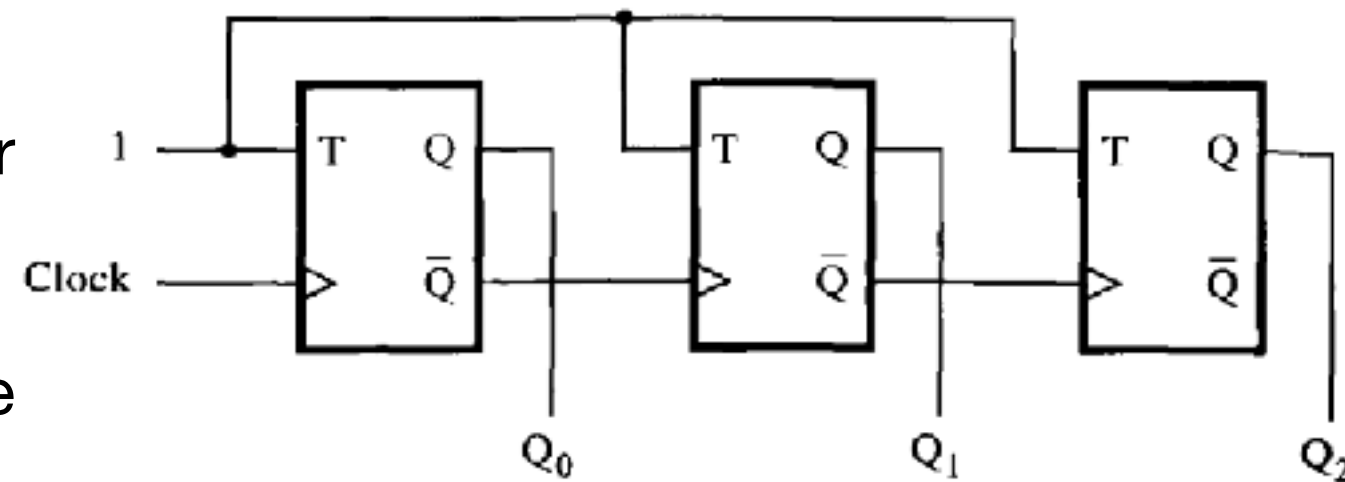
Counters

- Counter is a special purpose **adder/subtractor** which increments or decrement a **count by 1** for Up counter or Down counter respectively
- Counter circuits are used in digital systems for many purposes. They may count the number of occurrences of certain events, generate timing intervals for control of various tasks in a system, keep track of time elapsed between specific events, and so on.
- Counters can be implemented using the adder/subtractor circuits and the registers. However, since we only need to change the Contents of a counter by 1, it is not necessary to use such elaborate circuits.
- Instead, we can use much simpler circuits that have a significantly lower cost. Counter circuits can be designed using T and D flip-flops.

Asynchronous Counter

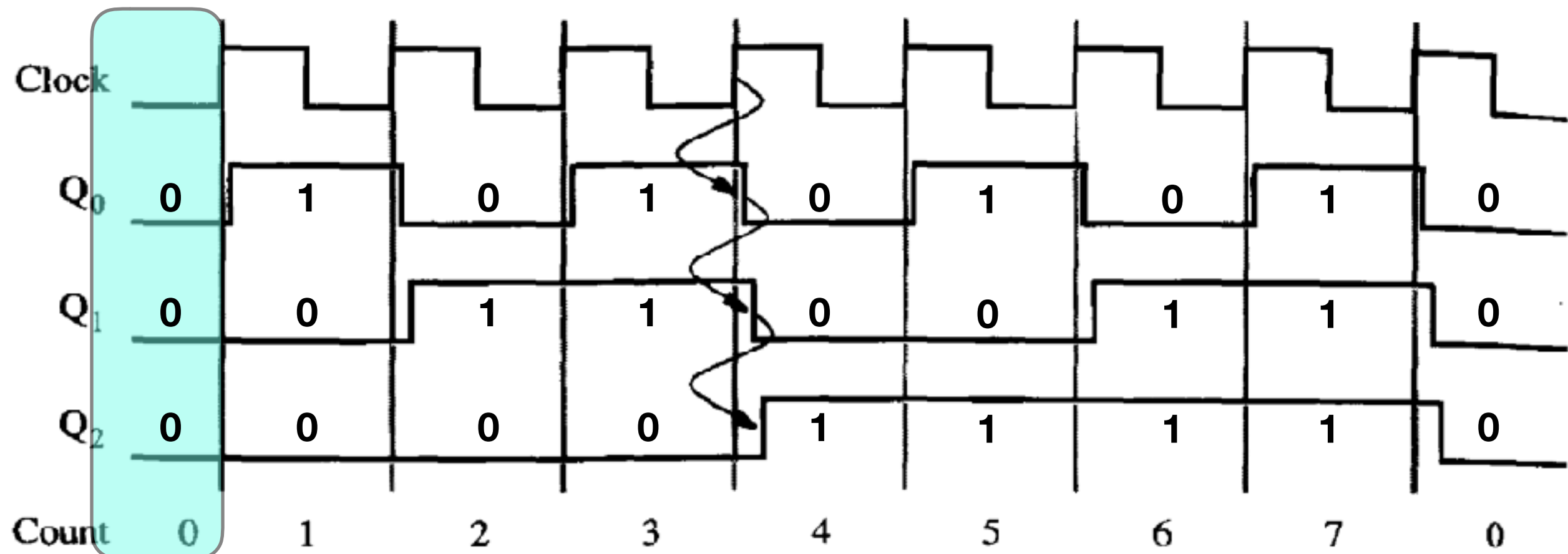
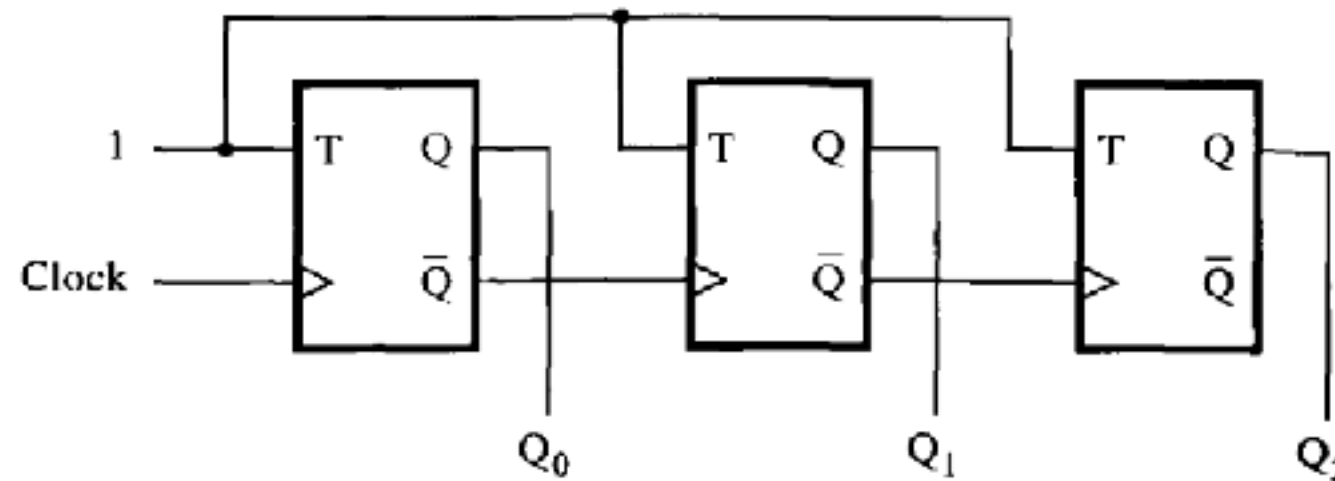
- Up Counter

- The simplest counter circuits can be built using T flip-flops because the toggle feature is naturally suited for the implementation of the counting operation.
- Shown figure is a three-bit counter capable of counting from 0 to 7.
- The clock inputs of the three flip-flops are connected in cascade.
- The T input of each flip-flop is connected to a constant 1, which means that the state of the flip-flop will be reversed (toggled) at each positive edge of its clock.



Asynchronous Counter

- Up Counter



Asynchronous Counter

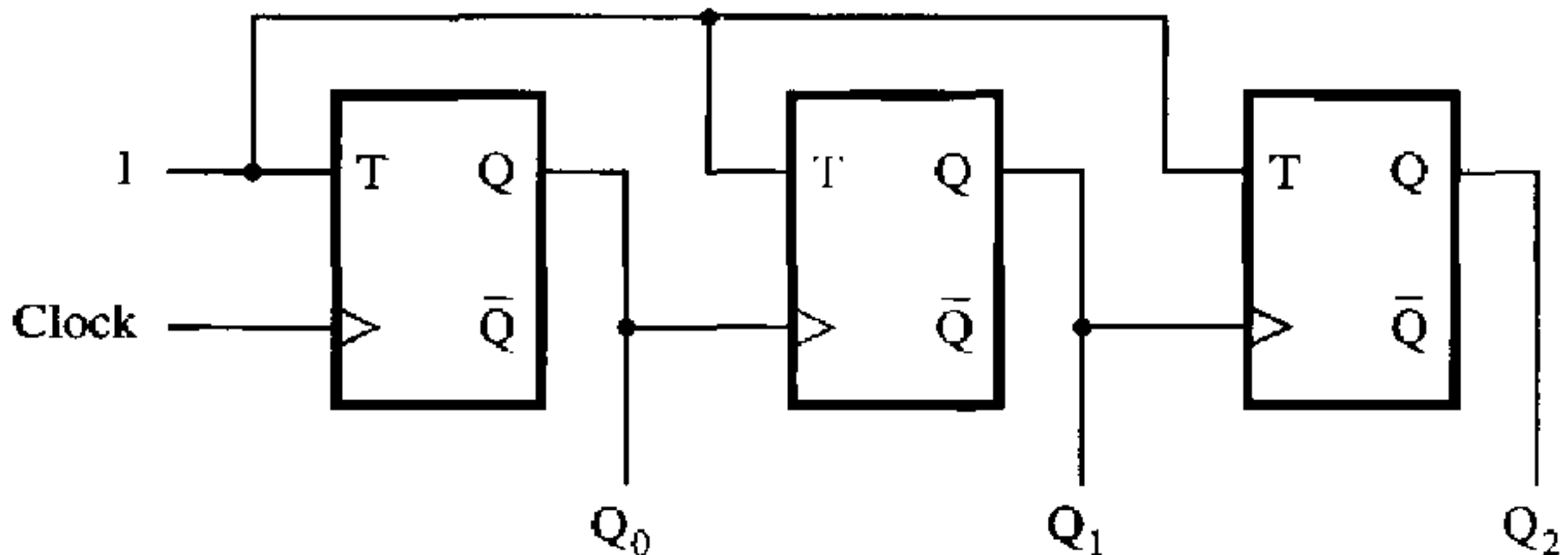
- Up Counter

- This circuit is a modulo-8 counter. Because it counts in the upward direction, we call it an up-counter.
- This counter has three stages, each comprising a single flip-flop. Only the first stage responds directly to the Clock signal; we say that this stage is synchronized to the clock.
- The other two stages respond after an additional delay. The change in Q1 appears after a second propagation, the change in Q2 occurs after a third delay.
- This behavior is similar to the rippling of carries in the ripple-carry adder circuit, hence it is called an asynchronous counter, or a ripple counter.

Asynchronous Counter

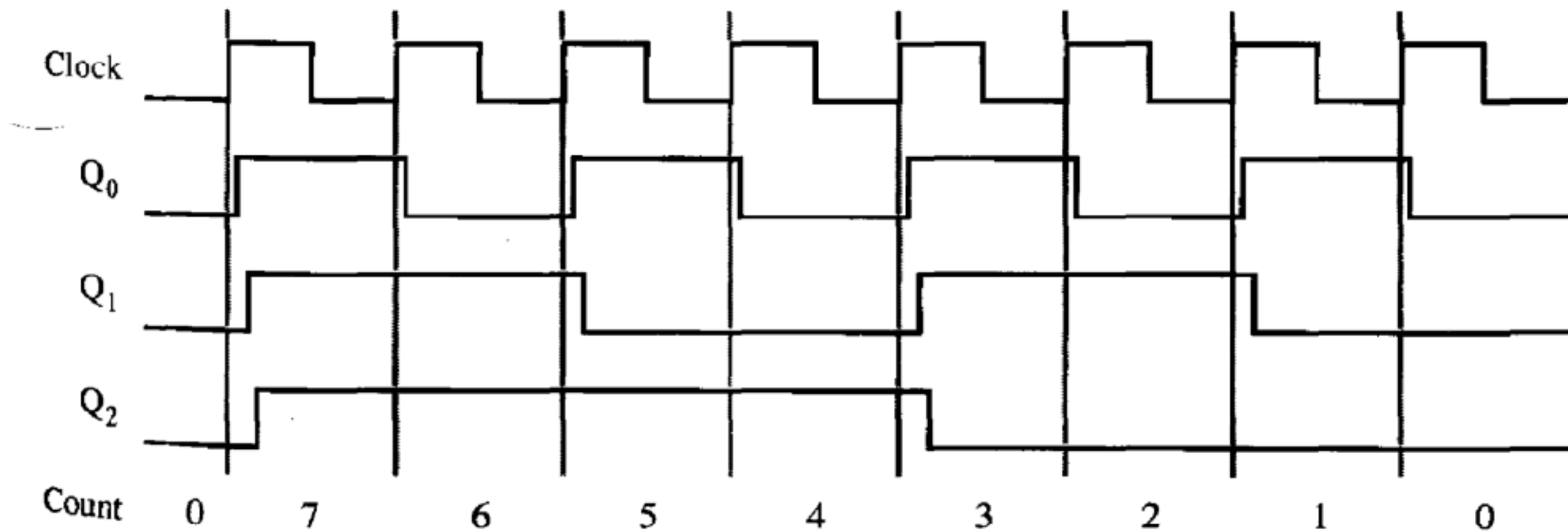
- Down Counter

- The only difference is that in the clock inputs of the second and third flip-flops are driven by the Q outputs of the preceding stages, rather than by the Q' outputs.



Asynchronous Counter

- Down Counter



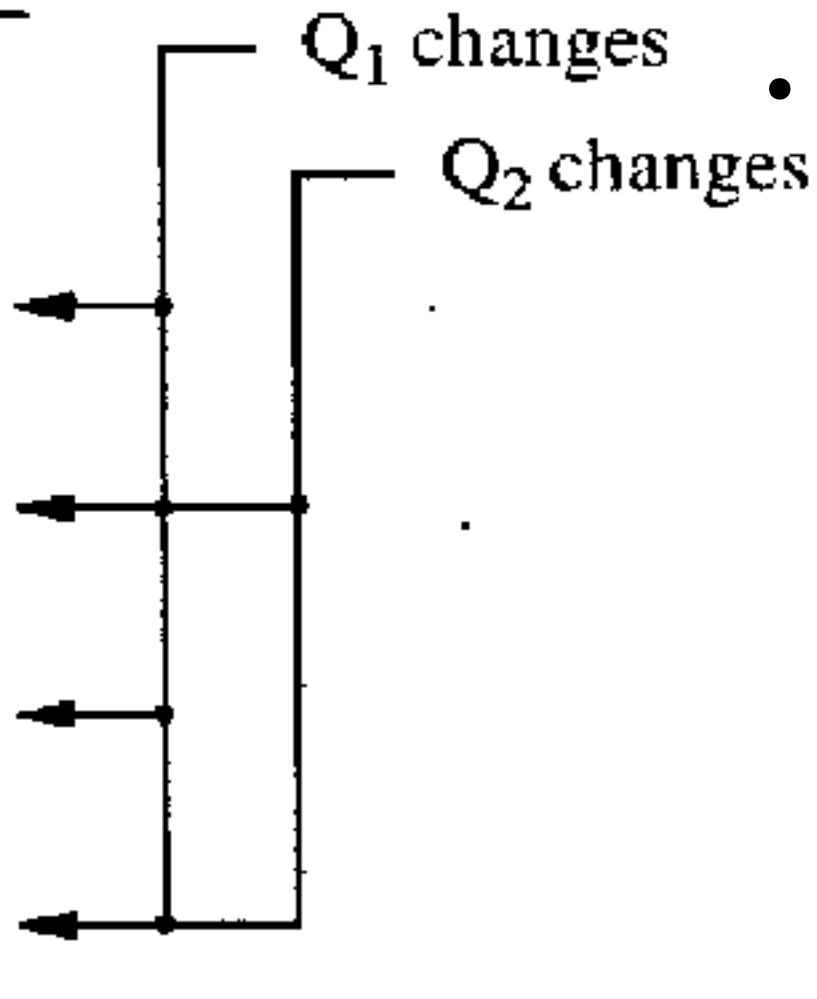
Synchronous Counter

- The asynchronous counters are simple, but not very fast
- If a counter with a larger number of bits is constructed in this manner, then the delays caused by the cascaded clocking scheme may become too long to meet the desired performance requirements.
- We can build a faster counter by clocking all flip-flops at the same time. This kind of counter is called as synchronous counter.

Synchronous Counter

- Q0 changes on each clock cycle
- Q1 changes only when Q0 = 1
- Q2 changes only when both Q1 and Q0 are equal to 1

Clock cycle	Q ₂	Q ₁	Q ₀
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0



$$T_0 = 1$$

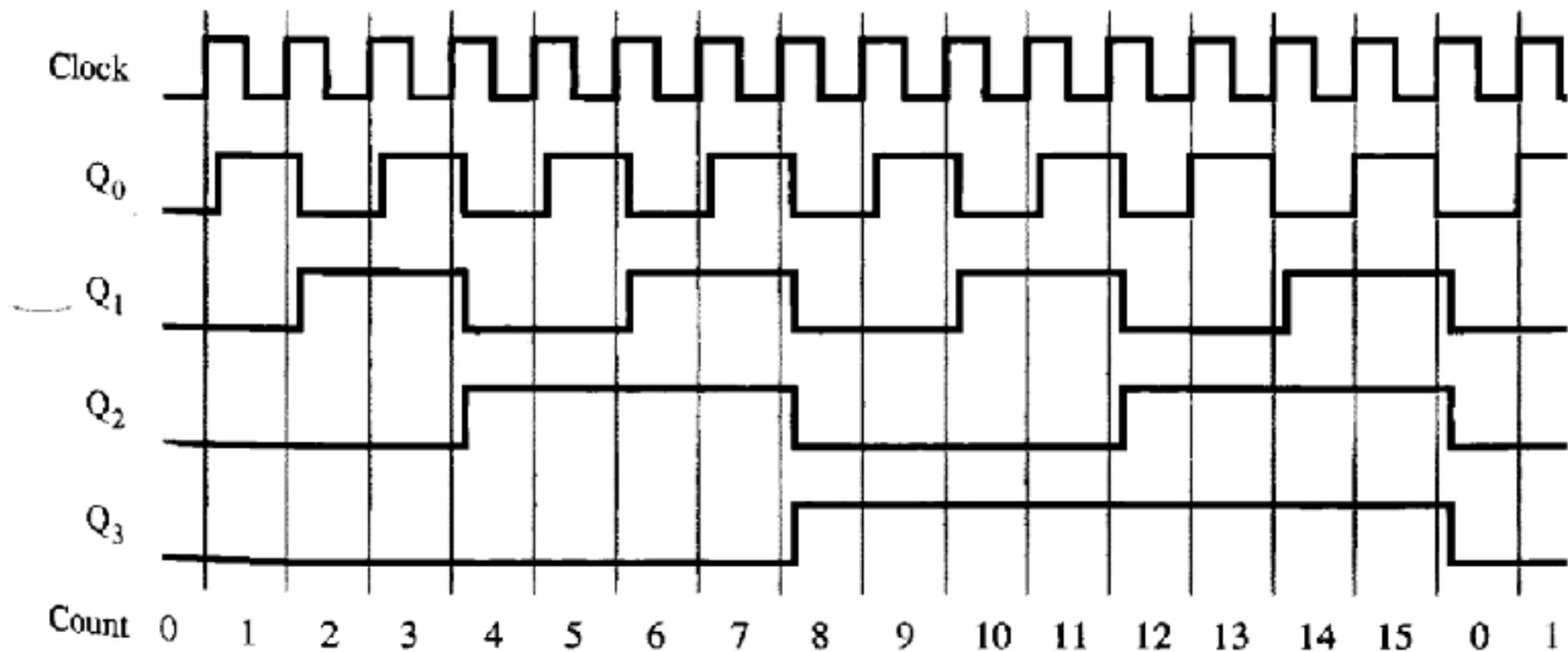
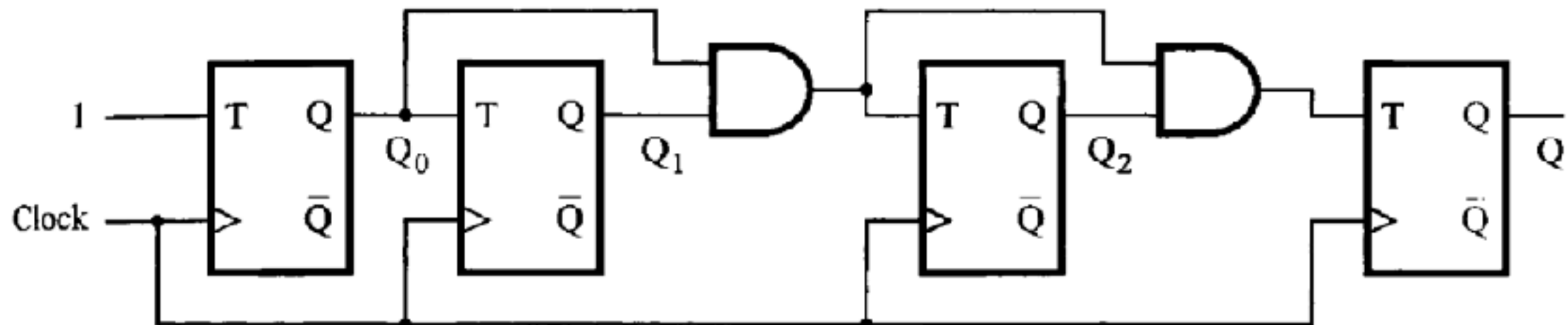
$$T_1 = Q_0$$

$$T_2 = Q_0 Q_1$$

$$T_3 = Q_0 Q_1 Q_2$$

$$T_n = Q_0 Q_1 \cdots Q_{n-1}$$

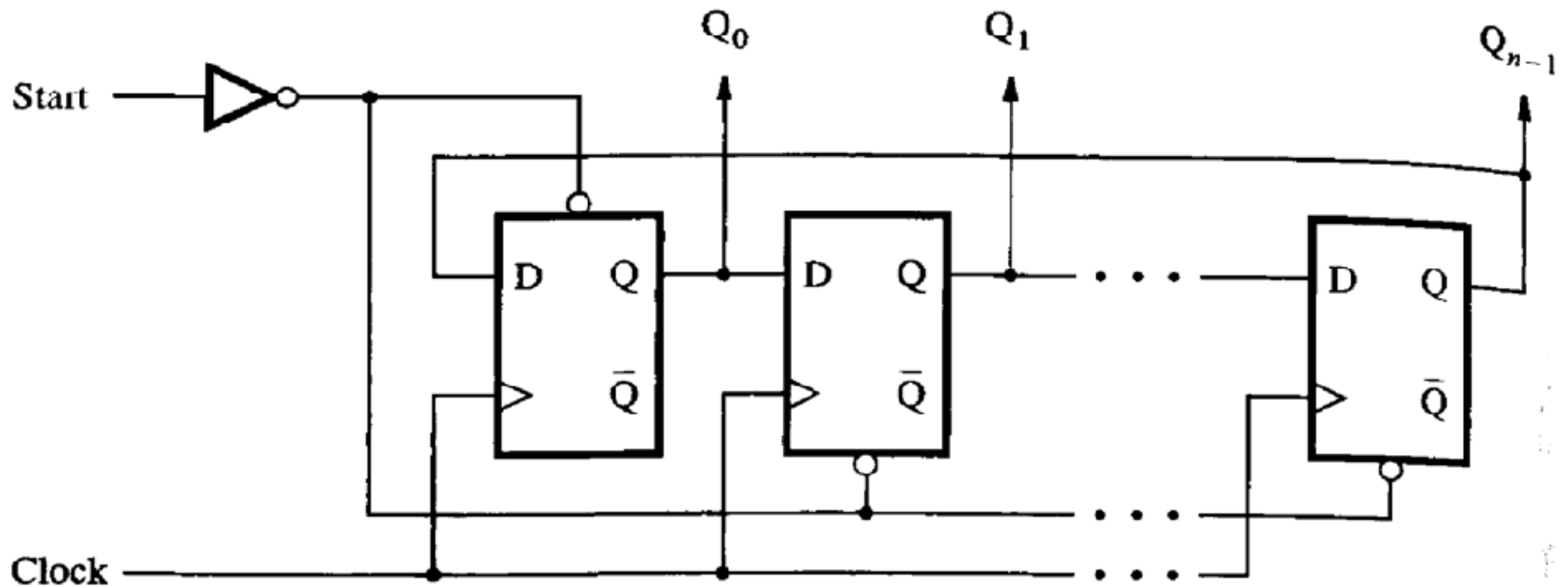
Synchronous Counter



Ring Counter

- It is possible to devise a counter like circuit in which each flip-flop reaches the state $Q_i == 1$ for exactly one count, while for all other counts $Q_i = 0$.
- Then Q_i indicates directly an occurrence of the corresponding count.
- Since this does not represent binary numbers, it is better to say that the outputs of the flips-flops represent a code.
- Such a circuit can be constructed from a simple shift register,

Ring Counter

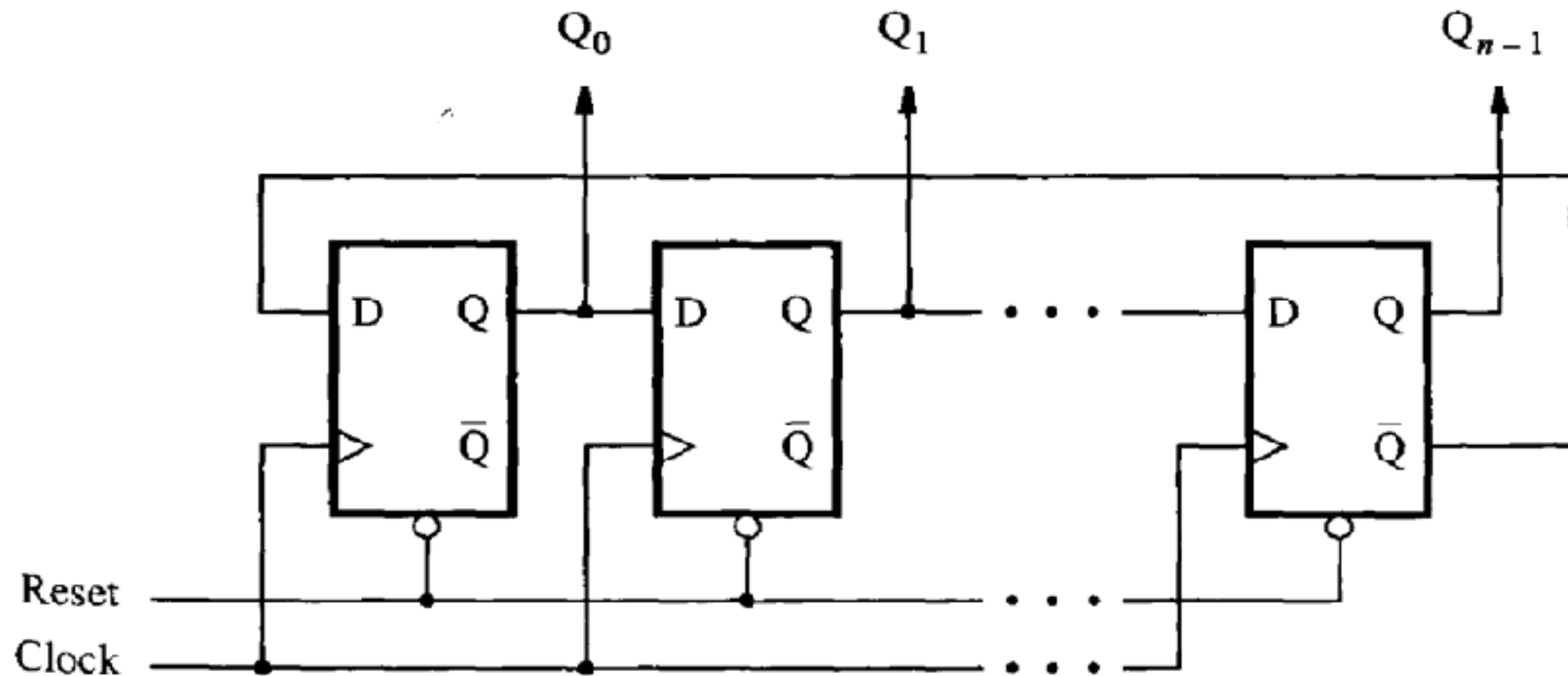


Ring Counter

- The Q output of the last stage in the shift register is fed back as the input to the first stage, which creates a ring like structure.
- If a single 1 is injected into the ring, this 1 will be shifted through the ring at successive clock cycles.
- For example, in a four-bit structure, the possible codes $Q_0Q_1Q_2Q_3$ will be 1000, 0100, 0010, and 0001.
- Such encoding, where there is a single 1 and the rest of the code variables are 0, is called a one-hot code

Johnson Counter

- An interesting variation of the ring counter is obtained if, instead of the Q output, we take the Q output of the last stage and feed it back to the first stage



Johnson Counter

- An n -bit counter of this type generates a counting sequence of length $2n$.
- For example, a four-bit counter produces the sequence 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, 0000, and so on.
- Note that in this sequence, only a single bit has a different value for two consecutive codes.

Thank You !!!